

SSMR / MAJOR SCIENTIFIC REVISION / 30 SEPTEMBER 2026

Diagnosing deterioration before choosing a feed action

Calibrated finite-grid diagnosis and physical action accounting in an ethanol membrane-reformer simulation

DRAFT - campaign results not released

This is a revised simulation study and evidence report. It is not a journal acceptance decision or a laboratory validation. The original submission and its historical reviews remain preserved.

What changed

The diagnostic now sees noisy, delayed measurements and accounts for their correlation. Its thresholds are calibrated before independent evaluation. The operating policy issues a real feed command, and both hydrogen shortfall and ethanol use are counted through the same horizon.

What succeeded and what failed

The methods and reporting structure are prepared. The campaign and independent checks are not yet complete, so diagnosis performance and action benefits are withheld. No success or failure of the revised scientific method is inferred from this draft.

Claims this evidence supports

A reproducible simulation study of a specific diagnostic experiment and feed policy, with explicit error rates, abstentions and physical trade-offs under declared assumptions.

Claims it does not support

Laboratory accuracy, a universally reliable diagnosis, an optimal feed experiment, economic payback or journal acceptance. More diagnostic information is useful only where the checked outcomes show that it changes an operating decision beneficially.

SCIENTIFIC REVISION / 01

What changed, and why

Revised abstract

A decline in reported hydrogen production can arise from process deterioration, impaired feed delivery or measurement bias. This simulation study compares passive acquisition with a scheduled feed move using delayed observations, joint covariance and a finite bank of evolving fault paths. A sequential exclusion rule is calibrated on separate observation episodes. Bounded feed policies are assessed by dynamic hydrogen and ethanol accounts. The campaign does not yet have a completed final verification record. Diagnostic performance, stress-test outcomes and action consequences are therefore withheld in this draft.

The original review found a gap between deterministic separation and statistical diagnosis. Reproducing the original arithmetic did not close that gap. The revised procedure is restricted to declared model, noise and policy assumptions.

Original concern	Revision implemented	Remaining boundary
Generating cause received a zero score	Fit every candidate to the same noisy observations	Performance is determined by held-out evaluation
Shared-baseline covariance was omitted	Use absolute window means and the full within-instrument covariance	Cross-instrument dependence remains a declared independence assumption
Threshold degrees of freedom were not justified	Calibrate maxima over the scheduled looks using finite-sample ranks	Protection applies only to covered finite truth strata and the stated noise law
Equal action names implied zero consequence difference	Issue an actual feed command and integrate physical outcomes	No service cost, price or monetary payoff is invented
A fixed-feed approximation supported the experiment claim	Simulate acquisition, ongoing deterioration and post-decision action	New evidence is Python simulation, not a new native or physical validation

Historical thresholds remain historical

The frozen original map and the later A1 amendment are prior evidence. Neither is presented here as a newly calibrated operating map. Their reproduction and chronology are retained in the preceding audit; the revised procedure has its own protocol, source freeze and checked outputs.

Active diagnosis is established methodology [3,4]. The candidate contribution is the bounded SSMR experiment and its action consequences. Whether it succeeds is an empirical result, not a premise of this document.

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Plant, candidate causes and dynamics

The physical truth trajectories use the staged benchmark equations and supplied initial state. Catalyst activity, membrane permeability and feed-delivery gain can evolve continuously during an episode. The diagnostic candidate bank uses evolving quasi-steady response templates; it is deliberately distinguished from the dynamic truth integration.

Label	Meaning	How it enters the episode
C	Catalyst deterioration	Activity follows the declared time history
M	Membrane deterioration	Permeability follows the declared time history
F	Feed-delivery loss	Actual ethanol differs from the commanded feed
S	Hydrogen-meter offset	The physical plant stays healthy; the hydrogen measurement is shifted
H	Healthy	Reference physical process and measurement behavior
CM	Combined deterioration	A stress case outside the single-cause candidate family

Initial state and fault timing

Before acquisition, the supplied state is settled at the scenario health at time zero while that health is held fixed. Failure to meet the declared settling criterion is retained as a computational failure. Deterioration then evolves inside the ODE right-hand side. A feed event changes the input without resetting the physical state.

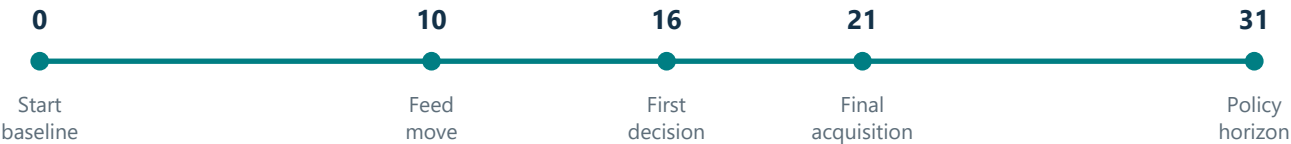
Scenario drop labels refer to a nominal quasi-steady map anchor at minute 10. They are not measured dynamic losses at that instant. Some anchors leave the declared response-map domain and are recorded as excluded before calibration. In particular, Mode 2 feed loss is investigated as an out-of-family stress; a map-domain restriction is not proof that feed loss is physically impossible.

The benchmark itself provides steady-state experimental model context [5,6]. This revision does not create new plant observations or validate evolving fault kinetics experimentally.

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Acquire observations before diagnosing

Acquisition clock, minutes; deterioration continues throughout



After a decision, branch from the saved physical state and account to the common horizon.

The active acquisition raises commanded ethanol by 0.0003 mol/min at minute 10; passive acquisition keeps the baseline command. Scheduled diagnostic looks occur at minutes 16, 21. All times refer to the acquisition clock, not elapsed time since a detected fault onset.

Sensor set	Available quantities
S1	Hydrogen production rate
S3	S1 plus outlet temperature and retentate volumetric flow
S4	S3 plus four gas-composition channels

Causal sampling and paired comparisons

Each instrument is sampled on its stated schedule. A chromatograph result is usable only after its reporting delay, even when its gas sample was acquired earlier. Window means are built from the actual samples available at the decision look. Reusing those same instrument records across sensor sets preserves the pairing of comparisons.

The two acquisition policies use matched noise draws for each scenario; calibration and evaluation occupy separate seed namespaces. The deterministic truth path can be reused across independent noise records. That produces repeated simulated observation episodes of one fixed plant scenario, not independently observed physical plants.

Timing denominator matters

Report a diagnosis time only for records that produced a singleton call. The final acquisition time is not an imputed diagnosis time for records that remain inconclusive or model-incompatible.

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A joint observation model

The analysis uses absolute means over the prespecified time windows. This avoids constructing an apparently independent baseline and difference from the same samples. Serial correlation, a persistent instrument bias, and reuse of samples are retained in the covariance of the summaries.

For instrument j , let A_j be the averaging matrix, K_j the covariance of its raw serial noise, and b_j the bound of its independent uniform record-level bias. The working covariance is:

$$C_j = A_j K_j A_j^T + (b_j^2/3) \mathbf{1}\mathbf{1}^T$$

The bias is drawn once per instrument and record. Its uncertainty does not vanish when more readings are averaged. Within-instrument errors follow the declared stationary AR(1) model; different instruments are independent in the nominal model. The full summary covariance is block diagonal across those instrument blocks.

What nuisance treatment means here

The candidate score profiles over its finite severity/rate templates. Persistent measurement bias is integrated through a declared random-effect covariance and calibration distribution; the procedure does not explicitly estimate every bias and does not guarantee validity for an arbitrary fixed bias within the bound. Gaussian innovations plus uniform bias also do not make the entire summary exactly Gaussian.

Instrument assumptions remain visible

The benchmark identifies an Agilent 3000 A MicroGC and a Bronkhorst hydrogen flowmeter, but does not supply the latter's exact model/range or the covariance and repeatability needed here. A membrane-surface K-type thermocouple is mentioned; it is not a specified outlet-temperature instrument. All diagnostic noise and delay inputs remain simulation assumptions [6,7].

A manufacturer accuracy limit is not automatically an independent Gaussian standard deviation. Calibration bias, repeatability, drift, gain error and response delay require separate interpretation [7].

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Calibration and sequential decisions

At each look, every supported cause is compared with the observed summary using the minimum full-covariance Mahalanobis distance over its frozen finite template bank. The diagnostic function receives observations and the bank, not the generating cause or future measurements.

$$D_{h,l}(z) = \min_{q \text{ in bank}(h)} (z - m_{h,l,q})^T C_l^{-1} (z - m_{h,l,q})$$

For each covered truth stratum, 499 independent calibration episodes provide the maximum true-cause score across both looks. With $\alpha = 0.05$, the threshold is the order statistic at rank $\text{ceil}((n + 1)(1 - \alpha))$. The cause-specific threshold is the maximum over its included truth-stratum thresholds. No chi-square reference distribution is imposed.

Decision condition	Recorded outcome
Exactly one cause remains at the first look	Stop and retain that singleton decision
Multiple causes remain	Continue to the final scheduled look
Exactly one remains at the final look	Report that singleton diagnosis
Multiple remain at the final look	INCONCLUSIVE
No cause remains	MODEL_INCOMPATIBLE

Membership is cumulative: a rejected cause cannot re-enter. The calibrated maximum accounts for the scheduled looks on the acquisition path. The subsequent physical policy begins at the recorded decision time and is accounted for separately.

Scope of the rank argument

The bound is marginal over fresh calibration and test draws for each included fixed truth stratum under the stated observation law. It is not a uniform claim over continuous severity, arbitrary bias, unseen kinetics or distribution shift. A particular realized calibration threshold can still have empirical error above the nominal target. [1,2]

Evaluation design and release criteria

Declared component	Protocol specification
Calibration	499 independent observation episodes per included calibration stratum
Nominal evaluation	1000 independent observation episodes per declared evaluation cell
Observation stresses	1000 episodes per selected stress cell; thresholds unchanged
Truth categories	covered grid: 61, heldout parameters: 13, kinetic mismatch: 4, faster deterioration: 4, out of family: 3
Scope of replication	Independent observation episodes conditional on the specified deterministic truth path

These are protocol specifications, not completion counts. The frozen checker must confirm actual coverage, split independence, causality, covariance, scores, thresholds, decisions and physical accounts before final publication of the results.

Outcome measures

For each fixed evaluation cell, report true-cause rejection, wrong singleton diagnosis, correct singleton diagnosis, inconclusive outcome and empty-set incompatibility. Every simulated record stays in the denominator. Exact pointwise binomial intervals describe uncertainty from the independent episode repetitions, not from serial samples within an episode.

Ranges across deterministic strata are descriptive ranges, not estimates of a population-wide rate. No scientific success is forced by the integrity gate: a reproduced false-rejection rate, poor power or unfavorable action trade-off can pass a computation check and still undermine the proposed procedure.

Separation of tuning and evaluation

batch_seed=20260930+case.index*100000+phase_offset; offsets
calibration0,evaluation10000,stress20000+stress_index*1000. Same draws paired between active/passive; independent across phases.

Failures under distribution shifts are retained. Changing thresholds or selecting favorable cases after viewing held-out results would require a new versioned protocol and a fresh evaluation.

Diagnosis results

Results withheld in this draft

This page will report checked per-stratum diagnosis outcomes for active and passive acquisition, including inconclusive and incompatible records. No provisional value is presented as a completed finding.

How to read these results

Correct calls measure useful resolution; a low wrong-call rate can coexist with frequent abstention. True-cause rejection is measured at the actual stopping look and is separate from the final outcome.

The healthy path overlaps the zero-fault limits of multiple candidate families. Healthy-case inconclusiveness is therefore an expected structural feature of this formulation, not a failure to hide by retuning thresholds.

Compare active/passive acquisition within the same scenario and sensor set. One favorable contrast cannot establish that a feed move is generally worth more than extra sensors. Passing arithmetic checks does not establish physical accuracy.

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What the diagnostic outcomes mean

Results withheld in this draft

This page will interpret the checked catalyst/membrane and healthy cases, including correct-call denominators and conditional timing. No provisional value is presented as a completed finding.

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Stress tests and failure boundaries

Results withheld in this draft

This page will report held-out parameter and model/noise stress results using unchanged thresholds. Failures remain part of the evidence. No provisional value is presented as a completed finding.

Prespecified departures from the nominal model

Observation stresses change noise scale, serial correlation, bias drift or multiplicative gain error. Physical stresses use off-grid parameters, altered kinetics, faster deterioration or simultaneous causes. The nominal covariance and thresholds stay unchanged so that loss of reliability remains visible.

The combined C/M and Mode 2 feed-loss cases lack a true label in the supported single-cause family. Their singleton calls therefore cannot be interpreted as correct cause identification. Reporting their incompatibility or abstention behavior is more informative than forcing an accuracy statistic with a nonexistent true candidate.

What a stress failure means

A failure identifies the boundary of the declared model and calibration. It does not invalidate arithmetic that the checker independently reproduces; conversely, numerical reproducibility does not excuse the failure. The report preserves both facts.

These selected stress cases are not exhaustive robustness evidence. Correlated instruments, unmodeled delays, plant-to-plant variability and real aging mechanisms remain possible departures from the assumptions.

Actions are physical commands

After diagnosis, the revised policy selects a bounded ethanol command rather than an action name. For each retained fitted path, it finds the smallest command on a fixed grid predicted to restore the nominal healthy map output at baseline feed through the common horizon. It then chooses the largest of those candidate-specific requirements.

If a retained candidate cannot meet the target on the grid, the policy uses the upper command bound and records infeasibility. An empty candidate set keeps the baseline command and records incompatibility. This is a declared conservative plug-in rule; it is neither an optimal controller nor a uniform safety guarantee.

A recorded pre-evaluation amendment distinguishes that restoration target from the benchmark's rounded demand: the original comparison could increase feed even for a zero-fault prediction. The replacement policy was frozen before calibration/evaluation; physical shortfall still uses the benchmark demand. The original protocol and helper remain preserved.

Physical account	Definition and interpretation
Hydrogen shortfall	Integral of $\max(\text{demand} - \text{true production}, 0)$, in mol
Hydrogen production	Integral of true production, in mol
Commanded ethanol	Integral of the commanded flow, in mol
Actual ethanol	Integral of delivered flow, in mol; differs under a feed-delivery fault
Active - passive difference	Signed paired contrast over the same truth, noise pairing and total horizon

Common horizon, continuous state

The accounts run from acquisition time zero to minute 31, including the diagnostic feed pulse. Each continuation begins from the exact saved ODE state at its decision time. Only the acquisition prefix before that decision is included; unused acquisition tails are excluded.

Command integrals follow the piecewise constant input intervals; true production and delivered-feed integrals use the declared endpoint quadrature within each segment. No integration interval bridges an unrepresented input discontinuity.

Policy consequences and interpretation

Results withheld in this draft

This page will report paired physical action outcomes from the common acquisition-plus-action horizon, with signed active-minus-passive differences. No provisional value is presented as a completed finding.

What can be concluded

Less hydrogen shortfall with more ethanol is a trade-off, not an automatic net benefit. Report the paired physical quantities and uncertainty together. Equal diagnosis action labels cannot force equal losses, and more diagnostic resolution does not guarantee an improved physical policy outcome.

Service downtime, recalibration success, membrane/catalyst replacement costs and the value of hydrogen or ethanol were not supplied. The revision therefore does not report monetary value of information, payback or a service recommendation justified by invented prices.

Policy limitation

The policy is selected from fitted candidate paths rather than a full uncertainty set over all health histories. A retained true cause does not ensure that its selected template predicts future production accurately. Performance must be read from the dynamic outcomes.

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Separate physical benefit from information value

The comparison does not isolate information value

The active experiment itself adds feed. Its hydrogen effect can arise directly from that extra feed as well as from a changed diagnosis or later command. No matched feed-only, nondiagnostic comparator was included. The checked differences measure the whole acquisition-and-action sequence; they cannot establish the separate causal value of diagnostic information.

Results withheld in this draft

This page will retain healthy and meter-bias physical consequences as well as deterioration cases. No provisional value is presented as a completed finding.

SCIENTIFIC REVISION / 13

Provenance, scope and next scientific step

Earlier native reproduction

The preceding full rerun reports successful numeric replay of the original study and retains machine-readable native MATLAB evidence. This closes the initial audit's missing-native-trajectory limitation for the covered baseline work. It does not establish statistical calibration of the old method or validate the new diagnosis procedure.

The native fine-grid record retained a diary/provenance limitation after disconnections. Source, state, solver-option and CSV checks were reported, while the absent diary was not silently converted into proof of execution provenance. A planned native repeat was not counted as completed evidence.

New method evidence

The revised acquisition, covariance, calibration and physical policy campaign is Python-based. It is a new computational experiment using staged reference equations, not another native MATLAB reproduction or a plant trial. The checker is an internal fresh implementation/review pass within the same model family, not independent physical or cross-model validation.

The remaining scientific boundary

The useful endpoint is a reproducible, honestly scoped computational result: where a cause can be resolved, where the rule must abstain, where assumptions fail, and what the issued feed command actually does. Application-specific evidence can support a simulation paper without implying deployability or journal acceptance.

Before a rig claim, identify the actual instruments and settings, measure joint repeatability and time dependence, characterize bias and delays, and test the complete diagnosis/action sequence against independently observed deterioration. Before a broad optimality claim, compare credible alternative excitations and policies under explicit constraints and losses.

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Primary references and source limits

[1] Dufour JM. Monte Carlo tests with nuisance parameters: A general approach to finite-sample inference and nonstandard asymptotics. *Journal of Econometrics*. 2006;133:443-477.

[doi:10.1016/j.jeconom.2005.06.007](https://doi.org/10.1016/j.jeconom.2005.06.007)

General rank/nuisance-test foundation; no direct SSMR guarantee.

[2] Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38:2074-2102.

[doi:10.1002/sim.8086](https://doi.org/10.1002/sim.8086)

Simulation design and Monte Carlo uncertainty.

[3] Scott JK, Findeisen R, Braatz RD, Raimondo DM. Input design for guaranteed fault diagnosis using zonotopes. *Automatica*. 2014;50:1580-1589.

[doi:10.1016/j.automatica.2014.03.016](https://doi.org/10.1016/j.automatica.2014.03.016)

Active input design for bounded linear model families.

[4] Raimondo DM, Marseglia GR, Braatz RD, Scott JK. Closed-loop input design for guaranteed fault diagnosis using set-valued observers. *Automatica*. 2016;74:107-117.

[doi:10.1016/j.automatica.2016.07.033](https://doi.org/10.1016/j.automatica.2016.07.033)

Closed-loop active diagnosis; different uncertainty assumptions.

[5] Serra M, Ocampo-Martinez C, Li M, Llorca J. Model predictive control for ethanol steam reformers with membrane separation. *International Journal of Hydrogen Energy*. 2017;42:1949-1961.

[doi:10.1016/j.ijhydene.2016.10.110](https://doi.org/10.1016/j.ijhydene.2016.10.110)

Membrane-reformer modeling and simulated control context.

[6] Arcila-Osorio M, Destro F, Ocampo-Martinez C, Llorca J, Braatz RD. A benchmark simulator for advanced control of ethanol steam reforming. *Renewable Energy*. 2026;256:124743.

[doi:10.1016/j.renene.2025.124743](https://doi.org/10.1016/j.renene.2025.124743)

Benchmark, steady-state validation and instrument description.

[7] JCGM. Evaluation of measurement data - Guide to the expression of uncertainty in measurement. JCGM 100:2008.

[doi:10.59161/JCGM100-2008E](https://doi.org/10.59161/JCGM100-2008E)

Interpretation of specifications, uncertainty and distribution assumptions.

A recent adjacent bibliographic lead is Santra, Koopman-PCE-based confidence-bounded fault diagnosis and recovery for hydrogen demand tracking in a membrane reactor, [doi:10.1016/j.compchemeng.2026.109728](https://doi.org/10.1016/j.compchemeng.2026.109728). Only publisher-indexed abstract/metadata were accessible; no full-method comparison or performance endorsement is made. See SOURCE_NOTES.txt for primary URLs, inspected sections and access limits.

SCIENTIFIC REVISION / APPENDIX A

Declared sensor and numerical assumptions

These settings are simulation inputs. Their display is not a claim that the named physical instruments attain these errors or time constants. SD denotes marginal per-sample random-error scale; the bias bound applies once per record.

Mo de	Quantity	SD	Bias bound	Period / delay (min)	AR(1)
1	H2 (mol/min)	1.137e-06	1.137e-06	0.1 / 0	0.6
1	Outlet T (K)	0.5	1	0.1 / 0	0.6
1	Retentate (m3/min)	2.858e-06	5.716e-06	0.1 / 0	0.6
1	H2 fraction	0.003968	0.007936	3 / 3	0.3
1	CH4 fraction	0.001	0.001336	3 / 3	0.3
1	CO fraction	0.001	0.0006267	3 / 3	0.3
1	CO2 fraction	0.001379	0.002758	3 / 3	0.3
2	H2 (mol/min)	1.382e-06	1.382e-06	0.1 / 0	0.6
2	Outlet T (K)	0.5	1	0.1 / 0	0.6
2	Retentate (m3/min)	1.905e-06	3.81e-06	0.1 / 0	0.6
2	H2 fraction	0.002744	0.005489	3 / 3	0.3
2	CH4 fraction	0.001	0.001683	3 / 3	0.3
2	CO fraction	0.001	0.0003748	3 / 3	0.3
2	CO2 fraction	0.001125	0.002251	3 / 3	0.3

Integrator and state treatment

Solver: BDF; axial discretization np = 50; relative tolerance 1e-06; absolute tolerance 1e-08; maximum step 0.1 min. Initialization and discontinuity handling follow the dynamic-engine contract.

Nominal model parameters are inherited from the staged public benchmark. These solver settings and repeatability checks are not a mesh-convergence demonstration. The run inventory retains raw trajectories, state snapshots, interval accounts, seed identities and intermediate diagnostic records.

SCIENTIFIC REVISION / APPENDIX B

Qualified computational verification

The final verification record is pending. A failed exact exported-start-state gate is being investigated by a separate frozen forensic replay. The original failed reports remain part of the evidence; this draft does not pre-accept a resolution.

What the internal checks can establish

The independent implementation checks recorded covariance, score and threshold arithmetic, seed separation, causal decisions, raw result summaries, and physical accounts. It is a fresh internal pass within the same model family. It is not a proof of the process equations, an experimental validation, or a cross-model peer review.

A qualified resolution does not alter the scientific performance findings. Wrong calls, inconclusive records, model incompatibility, stress failures and unfavorable feed costs remain reportable results. A passed integrity check cannot turn them into evidence of reliable plant diagnosis.

SCIENTIFIC REVISION / APPENDIX C

Audit trail and reproducibility

Release status

DRAFT: final results are withheld. missing checks/final_check.json

Files that define this document

Input	Purpose
protocol.json	Prespecified scenario, calibration, observation and policy rules
revision_model.py / dynamics.py	Implemented diagnosis and continuous physical truth/branching
POLICY_AMENDMENT.txt / revision_policy.py	Preserved pre-evaluation correction to the restoration target
results/diagnosis_summary.csv	Per-cell diagnostic counts, rates and uncertainty
results/policy_summary.csv	Paired physical policy outcomes
results/summary.json	Campaign completion and selected checked result summary
checks/final_check.json	Independent numerical/integrity gate
report/document_input_manifest.json	Exact input hashes and PDF/content hashes for this build

The PDF is generated from these inputs. A final build requires campaign completion, an explicit final checker acceptance with any qualification preserved, and matching current input hashes. The exact final-check file, result files and forensic bindings are recorded in the document manifest. Partial campaign numbers are not used to fill result pages. All pages are rendered for visual review and automatic text-bound checks.

Interpretation rules retained in every release

Pointwise episode-level intervals are not simultaneous guarantees across the scenario map. Descriptive ranges are not pooled population rates. No-call times are censored, not assigned a successful delay. Out-of-family faults have no supported true cause label. Physical costs are reported in their own units rather than compressed into an assumed price-weighted benefit.

The original archive, previous audit and complete rerun remain separate historical evidence. This revision does not rewrite their outcomes or manufacture a preregistration history. Source freezes demonstrate the recorded sequence only to the extent supported by the saved manifests and execution records.